

 **XTHLETE HACKATHON – PROBLEM STATEMENT**  
**Smart Fixture, Scheduling & Match Management System**

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 **Objective**

**Build a simplified but efficient system that can:**

- **Generate fair fixtures**
- **Avoid same-club matchups (as much as possible)**
- **Allocate byes correctly**
- **Schedule matches across multiple courts**
- **Ensure players/teams do not get overlapping matches**
- **Maintain relaxation time between matches**
- **Update results in real time**
- **Display standings, winners & runners**
- **Use secure match codes so only assigned umpires can submit results**

**Recommended stack: React/Next.js + Firebase + Node.js + Tailwind CSS**

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 **Core Features**

**1 Player, Team & Club Registration**

- **Register players with unique player ID**
- **Map each player to a club**
- **Players can register for multiple events (U15 singles, doubles, etc.)**
- **Prevent duplicate names or invalid entries**

**2 Fixture Generation**

**Supports formats:**

- **Knockout (mandatory)**
- **Round Robin (optional)**

**Must include:**

- **Same-club avoidance in early rounds**
- **Bye allocation for odd team counts**

**3 Smart Scheduling**

**The system must:**

- **Assign courts and time slots**
- **Ensure *no player has overlapping matches***
- **Ensure a *player receives minimum relaxation time* (e.g., 10 min for short events)**
- **If 3 courts start matches at 9:00 AM:**
  - **When Court 2 match ends at 9:10 AM,**
  - **Next match should start at 9:20 AM (10 min buffer)**
- **Keep courts active and avoid long idle periods**
- **Auto-adjust schedule if a match is delayed or a player withdraws**

**4 Match Code System for Umpires (Mandatory)**

**For every match:**

- **Generate a unique match code**
- **Store it in DB**

- Umpire must enter this code to:
  - ✓ Open the match scoring page
  - ✓ Submit or edit the score
- Umpire cannot access other matches without the corresponding code
- Code expires once match is completed

## 5 Results & Leaderboard

- Store match results in DB
- Auto-update winners & next-round fixtures
- Generate final standings
- Display live leaderboard on website

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## 🚫 Mandatory Constraints

### Player/Team Constraints

- Unique ID for each player
- A player cannot be assigned two matches at the same time
- Must have a minimum rest period between matches

### Club Constraints

- Avoid same-club matchups early

### Court Usage Constraints

- Multiple courts supported
- Minimize court idle time
- Add buffer time between consecutive matches

### Match Code Constraint

- Secure, unique match code per match
- Only umpires with valid code can update result

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## 🌱 Milestones

### Milestone 1: Player, Team & Club Module

- Player registration → unique player ID
- Team/club mapping
- Duplicate & invalid entry checks

#### Key Tests:

- ✓ Unique player ID generation
- ✓ Same club detection
- ✓ Duplicate entry rejection

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### Milestone 2: Fixture Engine

- Knockout fixture generation
- Same-club early-round avoidance
- Bye allocation

#### Key Tests:

- ✓ Bye correctness
- ✓ Same-club avoidance worked
- ✓ Correct match count

### Milestone 3: Scheduling + Results

- Multi-court scheduling
- No overlapping matches
- Relaxation time enforcement
- Court auto-allocation
- Result entry using match-code
- Auto-updated winner flow

#### Key Tests:

- ✓ No overlapping matches for any player
- ✓ Court idle time minimized
- ✓ Umpire cannot edit without valid match code
- ✓ Leaderboard updates correctly

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#### Evaluation Criteria

| Category                                    | Weight |
|---------------------------------------------|--------|
| Fixture Logic & Rules                       | 30%    |
| Scheduling Engine (Courts + Timings + Rest) | 30%    |
| Match Code Security & Result Handling       | 20%    |
| Code Quality & Architecture                 | 10%    |
| UI/UX & Presentation                        | 10%    |

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#### 🔗 Final Deliverables

1. Working Application / API
2. Documentation
  - Fixture logic
  - Scheduling logic
  - Match-code flow
  - DB schema
3. 3–5 min Video Demo
4. GitHub Repository

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#### End Goal

A simplified but powerful tournament engine that:

- ✓ Prevents scheduling conflicts
  - ✓ Uses match-level security
  - ✓ Keeps courts busy
  - ✓ Works across multiple events and categories
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